

A Three-Candidate Counterexample to Finite-Horizon Convexity in Robust Secretary Selection with Uncertain Acceptance

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Abstract

We consider the robust-ratio secretary problem with homogeneous uncertain acceptance probability p . Perez-Salazar, Singh, and Toriello empirically observed that the finite-horizon optimum $\gamma_n^*(p)$ appears to be convex and decreasing in p , and an associated open-problem formulation asks to establish or refute these shape properties for fixed n and for the limiting curve. This note gives a rigorous negative answer to the finite-horizon convexity part. For $n = 3$, we prove exactly that

$$\gamma_3^*(1/6) = \frac{546}{631}, \quad \gamma_3^*(11/60) = \frac{18421}{21600}, \quad \gamma_3^*(1/5) = \frac{21}{25}.$$

Since $11/60$ is the midpoint of $1/6$ and $1/5$, these values violate the midpoint inequality for convexity:

$$\gamma_3^*(11/60) - \frac{\gamma_3^*(1/6) + \gamma_3^*(1/5)}{2} = \frac{2419}{13629600} > 0.$$

Thus the finite-horizon convexity assertion is false already for three candidates. This is a solution, in the negative, of that subproblem; it is partial progress on the combined problem, because it does not settle monotonicity and does not settle the limiting curve $\gamma_\infty^*(p)$.

1 Introduction and scope of the result

In the secretary problem with uncertain acceptance (SP-UA), candidates arrive in a uniformly random order, the decision maker observes only relative ranks, and an offer is accepted independently with a fixed probability p . The performance measure of Perez-Salazar, Singh, and Toriello [1] is a robust ratio: the worst, over all k , of the probability of selecting a top- k candidate normalized by the probability that at least one top- k candidate would accept an offer. Their concluding remarks report numerical evidence that the optimal robust ratio $\gamma_n^*(p)$ is convex and decreasing as a function of p ; the open-problem formulation asks to establish or refute these properties for each fixed n and correspondingly for the limit $\gamma_\infty^*(p)$ [2].

The contribution of this note is deliberately narrow and exact. We give a three-candidate counterexample to finite-horizon convexity. Therefore the universal statement “ $\gamma_n^*(p)$ is convex for every fixed n ” is false. The calculation does not give a counterexample to monotonicity; the three values used below are in fact decreasing as p increases. Nor does a finite-horizon counterexample, by itself, decide convexity or monotonicity of $\gamma_\infty^*(p)$.

2 Model

There are n candidates with distinct absolute ranks $1 \prec 2 \prec \dots \prec n$, where rank 1 is best. The arrival order is a uniformly random permutation. At time t , the decision maker observes only the

partial rank of the current candidate among the first t arrivals, and then chooses whether to make an offer. If an offer is made, it is accepted with probability p , independently of all other randomness. Acceptance stops the process.

For a policy P and $k \in \{1, \dots, n\}$, define

$$D_k(p) = 1 - (1 - p)^k,$$

the probability that at least one of the top k candidates would accept an offer. The k th ratio of P is

$$\gamma_{P,k}(p) = \frac{\Pr(P \text{ selects a top-}k \text{ candidate and the offer is accepted})}{D_k(p)},$$

and the optimal robust ratio is

$$\gamma_n^*(p) = \sup_P \min_{1 \leq k \leq n} \gamma_{P,k}(p),$$

where the supremum is over admissible randomized policies. For the finite problem studied here the supremum is attained, because the feasible policy polytope is compact and the objective is continuous after introducing the usual minimum variable. The three-candidate reduction below also shows that Markov randomization suffices for the counterexample.

3 The exact three-candidate program

We now specialize to $n = 3$. At the last time step it is weakly optimal to make an offer: doing so cannot affect future decisions, and it can only increase the probability of selecting a top- k candidate for each k . Thus we restrict to policies that always offer at time 3.

Let

- $a \in [0, 1]$ be the probability of making an offer to the first candidate;
- x be the unconditional probability of reaching time 2, observing partial rank 1, and making an offer there;
- y be the unconditional probability of reaching time 2, observing partial rank 2, and making an offer there.

After the first candidate, the probability of reaching time 2 is $1 - pa$. Conditional on reaching time 2, the two partial ranks are equally likely. Consequently

$$0 \leq x \leq \frac{1 - pa}{2}, \quad 0 \leq y \leq \frac{1 - pa}{2}. \quad (1)$$

Conversely, every triple (a, x, y) satisfying (1) is implemented by offering in the time-2 partial-rank-1 state with probability $2x/(1 - pa)$ and in the time-2 partial-rank-2 state with probability $2y/(1 - pa)$, with the evident boundary convention when $1 - pa = 0$.

For a feasible triple (a, x, y) , write $G_k(a, x, y; p)$ for the resulting value of $\gamma_{P,k}(p)$. The next lemma gives the exact three-variable formulation.

Lemma 1 (Three-candidate formulation). *For $p \in (0, 1]$,*

$$G_1(a, x, y; p) = \frac{1 + (1 - p)a + (2 - p)x - py}{3}, \quad (2)$$

$$G_2(a, x, y; p) = \frac{2 + 2(1 - p)a + (3 - 2p)x + (1 - 2p)y}{3(2 - p)}, \quad (3)$$

$$G_3(a, x, y; p) = \frac{1 + (1 - p)(a + x + y)}{3 - 3p + p^2}. \quad (4)$$

Moreover,

$$\gamma_3^*(p) = \max_{\substack{0 \leq a \leq 1 \\ 0 \leq x, y \leq (1-pa)/2}} \min\{G_1(a, x, y; p), G_2(a, x, y; p), G_3(a, x, y; p)\}. \quad (5)$$

Proof. For any admissible policy, define a, x, y by the unconditional probabilities above. The bounds (1) are immediate from the reach probabilities at time 2. Conversely, every feasible triple is attained by the randomized Markov implementation described above. Thus it remains only to check that the three robust ratios depend on a policy through these three unconditional probabilities.

We compute the three normalized success probabilities. At time 1, the candidate is top k with probability $k/3$, so the contribution to the top- k success numerator is $pa k/3$.

At time 2, if the current candidate has partial rank 1, then among the three possible unordered pairs of first two arrivals the current candidate is absolute rank 1 with probability $2/3$ and absolute rank 2 with probability $1/3$. Hence its probabilities of belonging to the top 1, top 2, and top 3 sets are $2/3, 1$, and 1 , respectively. If the current candidate has partial rank 2, the corresponding probabilities are $0, 1/3$, and 1 . Thus the time-2 contributions to the top-1, top-2, and top-3 numerators are

$$p \cdot \frac{2x}{3}, \quad p \cdot \left(x + \frac{y}{3}\right), \quad p(x + y),$$

respectively.

Finally, the probability of reaching time 3 is $1 - p(a + x + y)$. For each fixed absolute rank $j \in \{1, 2, 3\}$, the probability of both reaching time 3 and having rank j appear last is

$$\frac{1 - p(a + x + y)}{3}.$$

Indeed, conditional on a fixed last rank, the second candidate has partial rank 1 in one of the two possible orders of the first two candidates and partial rank 2 in the other. Therefore the time-2 stopping probability conditional on that last rank is the same for all three last ranks. The time-3 contribution to the top- k numerator is consequently

$$p \cdot \frac{k}{3} (1 - p(a + x + y)).$$

Adding the appropriate time-1, time-2, and time-3 contributions and then dividing by

$$D_1(p) = p, \quad D_2(p) = p(2 - p), \quad D_3(p) = p(3 - 3p + p^2)$$

gives (2), (3), and (4). Equation (5) follows immediately. $\square$

4 Exact values at three probabilities

Lemma 2.

$$\gamma_3^*(1/6) = \frac{546}{631}.$$

Proof. First we give a feasible policy attaining the displayed value. Set

$$a = 1, \quad x = \frac{5}{12}, \quad y = \frac{61}{7572}.$$

For $p = 1/6$, the constraint bound is $(1 - pa)/2 = 5/12$, so the triple is feasible. Substitution into (2)–(4) gives

$$G_1 = \frac{546}{631}, \quad G_2 = \frac{18109}{20823} = \frac{546}{631} + \frac{91}{20823}, \quad G_3 = \frac{546}{631}.$$

Thus $\gamma_3^*(1/6) \geq 546/631$.

For the reverse inequality, let

$$H(p) = p^3 - 3p^2 + 3, \quad \lambda = \frac{3(1-p)}{H(p)}.$$

At $p = 1/6$ we have $\lambda = 540/631 \in [0, 1]$. The choice of λ cancels the coefficient of y in the convex combination $\lambda G_1 + (1 - \lambda)G_3$. Hence, for every feasible (a, x, y) ,

$$\begin{aligned} \min\{G_1, G_2, G_3\} &\leq \min\{G_1, G_3\} \\ &\leq \lambda G_1 + (1 - \lambda)G_3 \\ &= \frac{1 + (1 - p)a + 2(1 - p)x}{H(p)} \\ &\leq \frac{1 + (1 - p)a + (1 - p)(1 - pa)}{H(p)} \\ &= \frac{2 - p + (1 - p)^2 a}{H(p)} \\ &\leq \frac{p^2 - 3p + 3}{H(p)}. \end{aligned}$$

In the fourth line we used $x \leq (1 - pa)/2$, and in the final line we used $a \leq 1$. Evaluating the last expression at $p = 1/6$ gives $546/631$. Therefore $\gamma_3^*(1/6) \leq 546/631$, completing the proof. $\square$

Lemma 3.

$$\gamma_3^*(11/60) = \frac{18421}{21600}, \quad \gamma_3^*(1/5) = \frac{21}{25}.$$

Proof. Define

$$A(p) = \frac{p^2 - 5p + 6}{6}.$$

For any feasible (a, x, y) , using $y \geq 0$ and $x \leq (1 - pa)/2$ in (2) gives

$$\begin{aligned} G_1(a, x, y; p) &= \frac{1 + (1 - p)a + (2 - p)x - py}{3} \\ &\leq \frac{1 + (1 - p)a + (2 - p)x}{3} \\ &\leq \frac{1 + (1 - p)a + (2 - p)(1 - pa)/2}{3}. \end{aligned}$$

The final expression is affine in a , with coefficient

$$\frac{1 - 2p + p^2/2}{3}.$$

This coefficient is positive for $p = 11/60$ and for $p = 1/5$. Therefore, for these two values of p , the maximum of the preceding affine upper bound over $0 \leq a \leq 1$ occurs at $a = 1$, and

$$G_1(a, x, y; p) \leq \frac{p^2 - 5p + 6}{6} = A(p).$$

Consequently $\gamma_3^*(p) \leq A(p)$ for $p \in \{11/60, 1/5\}$.

Now take the feasible policy

$$a = 1, \quad x = \frac{1-p}{2}, \quad y = 0.$$

For this policy, $G_1 = A(p)$. Direct substitution into (2)–(4) gives

$$\begin{aligned} p = \frac{11}{60} : \quad G_1 &= \frac{18421}{21600}, \quad G_2 = \frac{16951}{19620}, \quad G_3 = \frac{15481}{17882}, \\ p = \frac{1}{5} : \quad G_1 &= \frac{21}{25}, \quad G_2 = \frac{116}{135}, \quad G_3 = \frac{53}{61}. \end{aligned}$$

In both displayed lines the values of G_2 and G_3 are strictly larger than G_1 . Thus the minimum value of $\{G_1, G_2, G_3\}$ is exactly $A(p)$ for each of the two stated values of p , matching the upper bound. $\square$

5 Failure of finite-horizon convexity

Theorem 4. *The function $p \mapsto \gamma_3^*(p)$ is not convex on $(0, 1]$. In particular, the finite-horizon assertion that $\gamma_n^*(p)$ is convex in p for every fixed n is false.*

Proof. Let

$$p_- = \frac{1}{6}, \quad p_0 = \frac{11}{60}, \quad p_+ = \frac{1}{5}.$$

Then $p_0 = (p_- + p_+)/2$. By Lemmas 2 and 3,

$$\begin{aligned} \gamma_3^*(p_0) - \frac{\gamma_3^*(p_-) + \gamma_3^*(p_+)}{2} &= \frac{18421}{21600} - \frac{1}{2} \left(\frac{546}{631} + \frac{21}{25} \right) \\ &= \frac{2419}{13629600} > 0. \end{aligned}$$

A convex function must satisfy

$$f\left(\frac{p_- + p_+}{2}\right) \leq \frac{f(p_-) + f(p_+)}{2}.$$

The displayed strict reverse inequality proves nonconvexity. $\square$

Corollary 5. *The combined finite-horizon statement that $\gamma_n^*(p)$ is both convex and decreasing for every fixed n is false, because its convexity component already fails at $n = 3$.*

Remark 6 (What is, and is not, solved). Theorem 4 is a complete negative solution of the finite-horizon convexity subproblem. It is only partial progress on the full problem stated in [2]. The proof does not disprove monotonicity of $\gamma_n^*(p)$; indeed the three exact values used above satisfy

$$\frac{546}{631} > \frac{18421}{21600} > \frac{21}{25}.$$

The proof also does not settle whether the limiting curve $\gamma_\infty^*(p)$ is convex or decreasing.

References

- [1] Sebastian Perez-Salazar, Mohit Singh, and Alejandro Toriello. Robust online selection with uncertain offer acceptance. *Mathematics of Operations Research*, 50(3):2226–2260, 2025. doi: 10.1287/moor.2023.0210.
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